

# DHYEY VIPULBHAI DESAI

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## PROFILE SUMMARY

AI/ML engineer building agentic AI workflows, end-to-end RAG pipelines, and LLM evaluation harnesses for enterprise clients (MS Applied Data Science, USC). Skilled in LangGraph, hybrid retrieval and reranking, multimodal ML, prompt engineering, and deployment on AWS and Spark.

## EDUCATION

### University of Southern California

Master of Science, Applied Data Science | GPA: 3.73

Aug 2024 - May 2026

United States of America

- **Coursework:** Foundations of Data Management, Machine Learning for Data Science, Foundations and Applications of Data Mining

### Manipal University Jaipur, Rajasthan

Bachelor of Technology, Computer Science and Engineering | GPA: 3.85

Jul 2020 - May 2024

India

- **Coursework:** Data Science and Machine Learning, Image Processing and Pattern Analysis, Artificial Intelligence, Regression Analysis

## PROFESSIONAL EXPERIENCE

### Associate Forward Deployed Engineer

Momentum Blue (A Coforge Company) | Cross-Industry Enterprise AI

Aug 2026 - Present

Princeton, New Jersey

- **Enterprise Agentic AI Delivery Readiness:** building deployment-ready capability to design, evaluate, and deliver enterprise agentic AI.
- Built LangGraph agentic workflows and end-to-end RAG pipelines (ChromaDB, pgvector, BM25 hybrid retrieval, Cohere reranking) to Momentum Blue delivery standards, with a fixed evaluation harness for retrieval recall and answer quality.
- Ran LLM evaluations for agent and RAG quality and packaged reusable solution patterns for client deployments; a policy RAG harness scored 1.0 retrieval recall and answer accuracy on 16 questions.

### Data Science Co-op

Starcycle | LegalTech / Corporate Dissolution Platform

Feb 2026 - May 2026

United States of America

- Prototyped Python tools and ETL pipelines parsing 1K+ inputs to automate entity and tax data enrichment for wind-down compliance.
- Built late-fee and closure indexes, reconciled records across internal and third-party systems, and generated synthetic and knowledge-graph-backed datasets for 5+ dashboards that surfaced platform and funnel bottlenecks.
- Designed and evaluated 50+ prompts, context strategies, and API/MCP integrations to improve internal AI tool quality and reliability.

### AI/ML Engineer Intern

Onawa Pet | PetTech | [GitHub](#)

May 2025 - Aug 2025

United States of America

- Built a RAG system (OpenAI function calling, Firebase) over 500+ voice notes, delivering pet health analytics across 3 dashboards.
- Built a TensorFlow/Transformers pipeline for 200+ audio files (transcription, PDF analysis, summarization), cutting manual review by 60%.

### Generative AI Engineer Intern

Genpact | IT Consulting / Digital Transformation

Apr 2024 - Jun 2024

India

- Built a conversational AI assistant for purchase-order automation with multi-document querying over large purchase orders.
- Designed and optimized GPT-4 prompt sets with custom evaluation metrics for accuracy, relevance, and robustness, improving query precision.

## PROJECTS

### Policy RAG Assistant | [GitHub](#)

- RAG assistant over policy documents using Gemma, Chroma, and Cohere reranking; 1.0 retrieval recall and answer accuracy on 16 questions.

### BEACON | Emergency Decision Support for First Responders | [GitHub](#)

- QLoRA fine-tune of Gemma 4 E4B on WHO/SPHERE/IMCI protocols; offline BM25 RAG, voice/photo/text input, FastAPI on Docker.

### MultiLLM | Intelligent Multi-Model AI System

Sep 2025 - Dec 2025

- Privacy-first platform routing tasks across 5+ local Ollama models (Llama 3.2, DeepSeek Coder, Phi3) with streaming chat and Google OAuth.

## TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, C, Scala | **AI Coding Tools:** Cursor, Claude, GitHub Copilot
- **ML & GenAI:** LangGraph, RAG, Hybrid Retrieval (BM25), Reranking, Function Calling, QLoRA, Transformers, Recommender Systems
- **LLM Evaluation:** Prompt Evaluation, Citation Checks, Fixed-Set Recall Tests | **Multimodal & NLP:** CV, NLP, NER, Text-to-SQL
- **ML Engineering:** FastAPI, Pydantic, REST APIs, Docker, GitLab CI, pytest, Ruff, Mypy, Spark RDD, Hadoop, ETL, Quantization
- **Tools & APIs:** Hugging Face, Ollama, TensorFlow, OpenAI/Gemini/Cohere APIs, ChromaDB, pgvector, MongoDB, Databricks, Git, Tableau
- **Cloud:** AWS Certified AI Practitioner (2026), AWS Certified Cloud Practitioner (2025)

## PUBLICATIONS

- **Amplified Silence: Uncovering and Mitigating Gender Bias in Music Recommendation Algorithms** – accepted and presented at the 4th Music Recommender Systems Workshop (MuRS 2026), held with RecSys, Sept 2026. Stacking a post-processing boost with Fair-ALS cut male-artist exposure from 69.8% to ~50% at 30% lower NDCG cost than post-processing alone.
- **Pseudo-RGB Slice Stacking in a 2D ResUNet for High-Sensitivity Multiple Sclerosis Lesion Segmentation, Diagnostics (MDPI)** ([code](#)).
- **Supervised Machine Learning Approaches for Brain Stroke Detection**, IEEE IEMECON 2023: logistic regression reached 96% accuracy.

## ACHIEVEMENTS

- **Origin Weekend: IMPACT S26 – First Place:** won a startup launch sprint organised by USC, Google, and TIE Hub (USC Viterbi).
- **Graduate Teaching Assistant, DSCI 551 & DSCI 351:** supported 100+ students via weekly sections; designed assignments with faculty.
- **AWS re:Invent:** All Builders Welcome Grant recipient (2025); Alumni Advisor (2026) | **Dean's List** (2023)
- **Advanced Deep Learning Research, NUS:** privacy-preserving facial analysis; +10% accuracy, 40% lower latency on 3× larger batches.